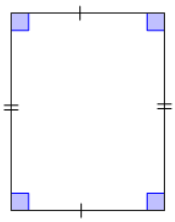
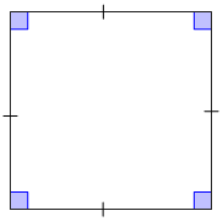
Classifying quadrilaterals

1 Give the most specific classification of the following quadrilaterals as trapezoid, rectangle, or square:

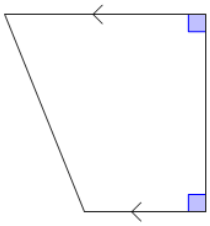
a



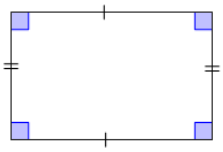
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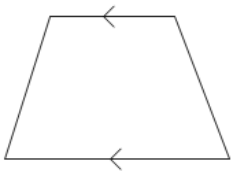
c



d

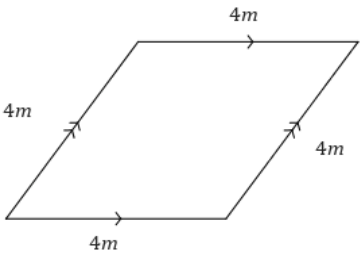


e



2 Give the most specific classification of the following quadrilaterals as parallelogram, trapezoid, or rhombus:

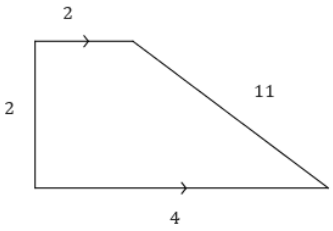
a



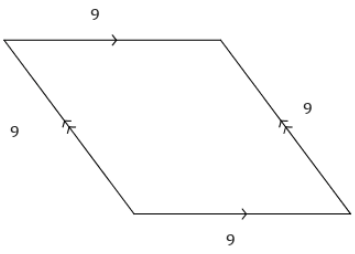
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c

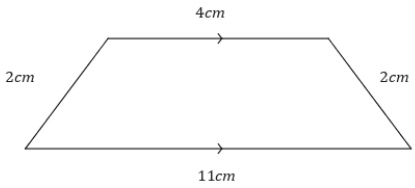


d

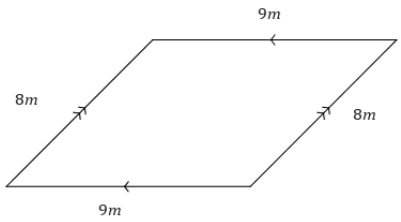


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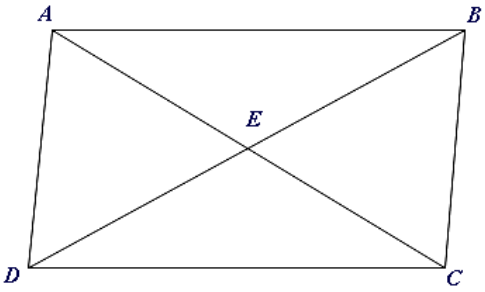
e



f



3 In the given figure, $\triangle AEB \cong \triangle CED$.
What type of quadrilateral is $ABCD$?



4 Determine whether each statement is true or false.

a A trapezoid is a parallelogram.

b A square is a rhombus.

c A parallelogram is a rectangle.

d A kite is a rhombus.

e A rhombus is a square.

f All parallelograms are quadrilaterals.

g All quadrilaterals are parallelograms.

h All rhombi have a pair of parallel sides.

i All quadrilaterals with a pair of parallel sides are rhombi.

j All squares are parallelograms.

k All rhombi have congruent consecutive sides.

l All parallelograms are squares.

m All quadrilaterals with a a pair of congruent consecutive sides are rectangles.

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5 For each of the following scenarios, state the most specific classification of the quadrilateral.

a Patricia draws a quadrilateral, and covers it up. She tells Glen that the quadrilateral consists of right angles only.

b Hannah draws a parallelogram, and covers it up. She tells Glen that the opposite sides are parallel.

c Aaron draws a parallelogram, and covers it up. He tells Kathleen that consecutive sides are congruent and the diagonals are perpendicular.

d Marge draws a parallelogram, and covers it up. She tells Peter that the figure has all right angles.

e Marge draws a parallelogram, and covers it up. She tells Paul that the diagonals of the parallelogram are perpendicular.

f Bob draws a parallelogram, and covers it up. He tells Emma that the diagonals bisect each other.

g Tom draws a parallelogram, and covers it up. He tells Rosey that consecutive sides are congruent.

h Michael draws a parallelogram, and covers it up. He tells Xanthe that consecutive angles are supplementary.

6 For each of the following descriptions, state the most specific classification of the quadrilateral.

a A quadrilateral with opposite sides that are congruent.

b A quadrilateral with all angles congruent.

7 For each of the following descriptions, give an additional information to guarantee that the quadrilateral is a parallelogram.

a A quadrilateral with one pair of opposite sides that are parallel.

b A quadrilateral with perpendicular diagonals.

c A quadrilateral with one diagonal bisects the other.

8 Determine what must be shown to prove that the parallelogram is:

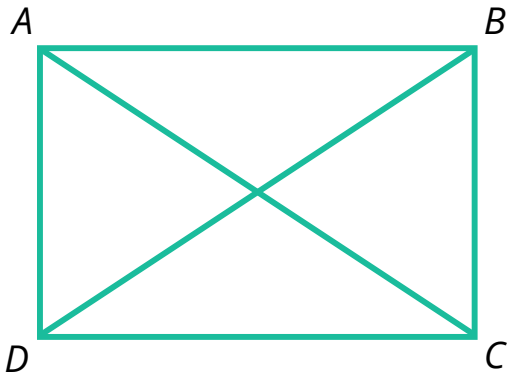
a Rhombus **b** Rectangle **c** Square

9 Determine what must be shown to prove that a given trapezoid is isosceles in terms of:

a Diagonals **b** Base angles **c** Opposite angles

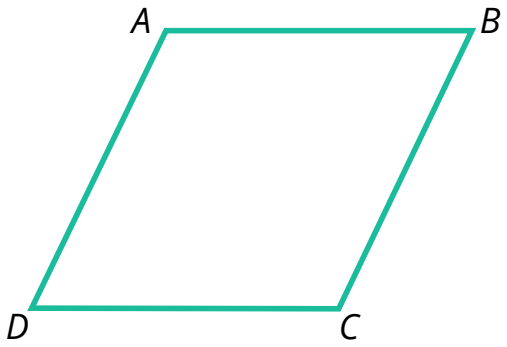
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11 Given that $ABCD$ is a parallelogram, and $\overline{AC} \cong \overline{BD}$, prove that $ABCD$ is a rectangle.

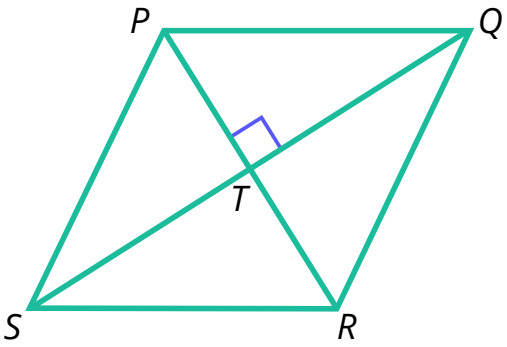


12 Prove that the following is a rhombus with the given conditions:

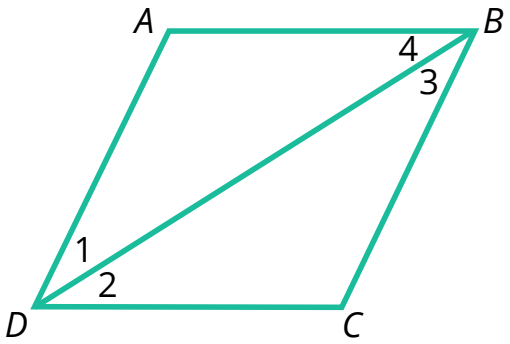
a Given that $ABCD$ is a parallelogram, and $\overline{AB} \cong \overline{BC}$.



b Given that $PQRS$ is a parallelogram, and $\overline{PR} \perp \overline{QS}$.



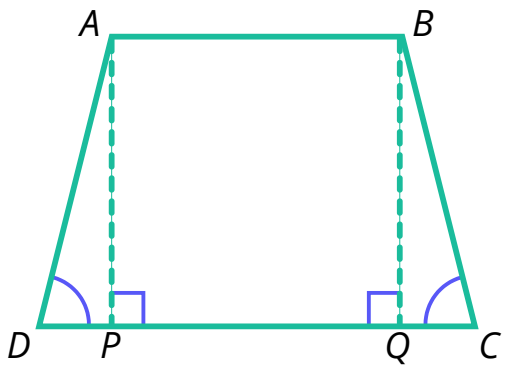
c Given that $ABCD$ is a parallelogram, and $\overline{BD}$ bisects $\angle ABC$ and $\angle ADC$.



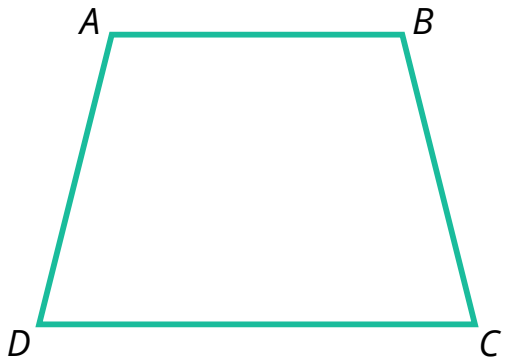
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13 Prove that the following is an isosceles trapezoid with the given conditions:

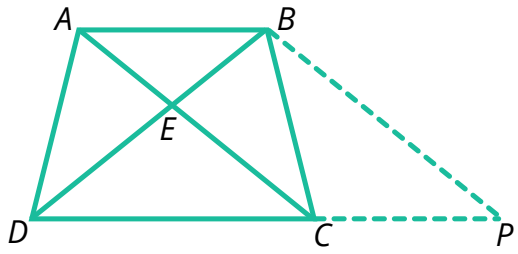
a Given that $ABCD$ is a trapezoid with altitudes $\overline{AP}$ and $\overline{BQ}$ and that $\angle ADC \cong \angle BCD$.



b Given that $ABCD$ is a trapezoid such that $\angle A$ and $\angle C$ are supplementary.



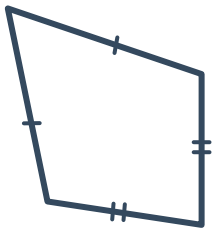
c Given that $ABCD$ is a trapezoid such that $\overline{AC} \cong \overline{BD}$, and $\overline{AC} \parallel \overline{BP}$.



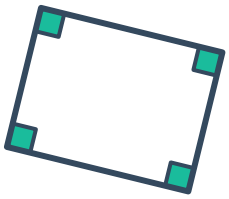
Additional questions

14 Classify the following quadrilaterals using all the terms that apply out of: rectangle, rhombus, kite, parallelogram, trapezoid, or neither.

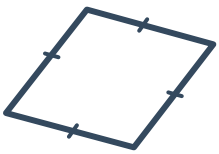
a



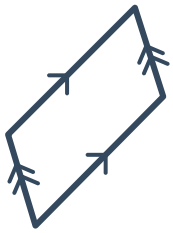
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c

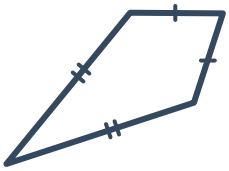


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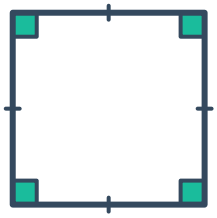


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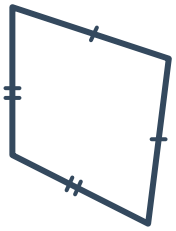
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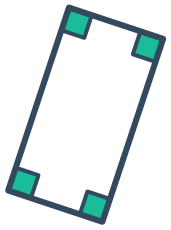
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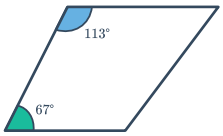
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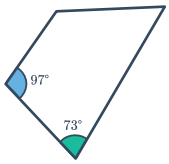
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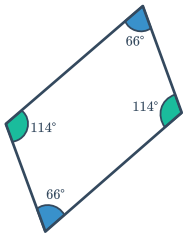
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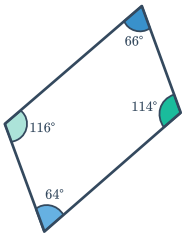
k



l



m



15 What is the largest number of obtuse angles that a quadrilateral can have? Explain your answer.

16 What is the largest number of right angles that a quadrilateral can have? Explain your answer.

17 Can a quadrilateral have exactly 2 right angles? If so, name a quadrilateral that could have two right-angles.

18 Can a quadrilateral have exactly 3 right angles?

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